Sahaj Bhadja

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Summary

Software Engineer with 2+ years building business applications, **SQL** database integrations, and reporting in Python, Java, **C#**, and TypeScript, with test plans, **root cause analysis**, documentation, and client-facing delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Data Retrieval, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built **data integration** pipelines between FastAPI services, SQL Server, PostgreSQL, and cloud storage with incremental loads and reconciliation checks, running 5K+ monthly jobs at 99%+ reliability through CI/CD releases.
- Designed normalized **relational schemas** with indexing for 200+ users and 10K+ records, tuning SQL to cut query latency 35% and dashboard reporting latency from 1.2s to 350ms.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed a web application with a React frontend and Python Flask services backed by MySQL on AWS for upload, rubric, and grading workflows, serving 150+ instructors and 10K+ daily requests.
- **Troubleshot** production failures by tracing OCR timeouts to their root cause and adding fallback routing and timeout handling, keeping APIs stable at peak load and cutting feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Took data-sharing software from **requirements analysis** through deployment with Microsoft and GitHub stakeholders, writing and executing 20+ unit and integration test suites run nightly with CSV result reports.
- Redesigned ETL on AWS Glue and S3 that consolidated activity data from multiple sources into normalized formats for **reporting**, boosting pipeline efficiency 68% and improving API response times 30%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **C# .NET** resource-management application on Azure SQL with REST APIs for resource allocation, invoices, payments, and leads, syncing 2M+ records across multiple sites in real time.
- Modeled Azure SQL tables for usage statistics and **performance reports**, and shipped through Jenkins CI/CD with automated tests and staged releases, helping cut resource wastage 42% through analytics.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Gathered requirements with the client, mapped 12+ integration points, and **presented the system architecture** before implementation, then built booking and payment APIs with responses under 200ms across 50K+ records.
- Built hourly Spark analytics on Databricks over 50K+ booking and payment events, producing revenue, demand, and active-user **reports** for internal dashboards, operations reviews, and recurring client reporting needs.

Skills

Languages: C#, Python, Java, JavaScript, TypeScript, SQL, HTML, CSS
Databases: SQL Server, Azure SQL, PostgreSQL, MySQL, Database Design, Query Optimization, Indexing
Applications: .NET, FastAPI, Flask, React, REST APIs, Microservices, Data Integration, ETL Pipelines
Reporting & Data: Spark, Databricks, Tableau, Dashboards, AWS Glue, Azure, AWS, Jenkins, Agile
Quality & Delivery: Test Plans, Unit Testing, Integration Testing, PyTest, Debugging, Root Cause Analysis, Documentation, Git

Projects

Plant Production Reporting Dashboard | *C#, ASP.NET Core, SQL Server, React, Chart.js*

- Designed a C# and React dashboard that pulls shift production counts, downtime events, and scrap rates from a **SQL Server** historian into daily and weekly reports with filterable **OEE** trends by line.

Batch Record Integration Service | *Python, OPC UA, SQL Server, REST APIs, PyTest*

- Designed a Python service that reads batch start, end, and quality values from plant equipment over **OPC UA**, validates them, and posts completed batch records to an **ERP** system through REST APIs.

Chat2SQL Code | *Java, TypeScript, SQL Server, PostgreSQL, MySQL, Docker*

- Extended a database workspace supporting 10+ engines including **SQL Server**, with schema synchronization, a query console, AI-assisted reporting dashboards, and AI-generated SQL reaching 82% accuracy across Windows, macOS, and Linux.